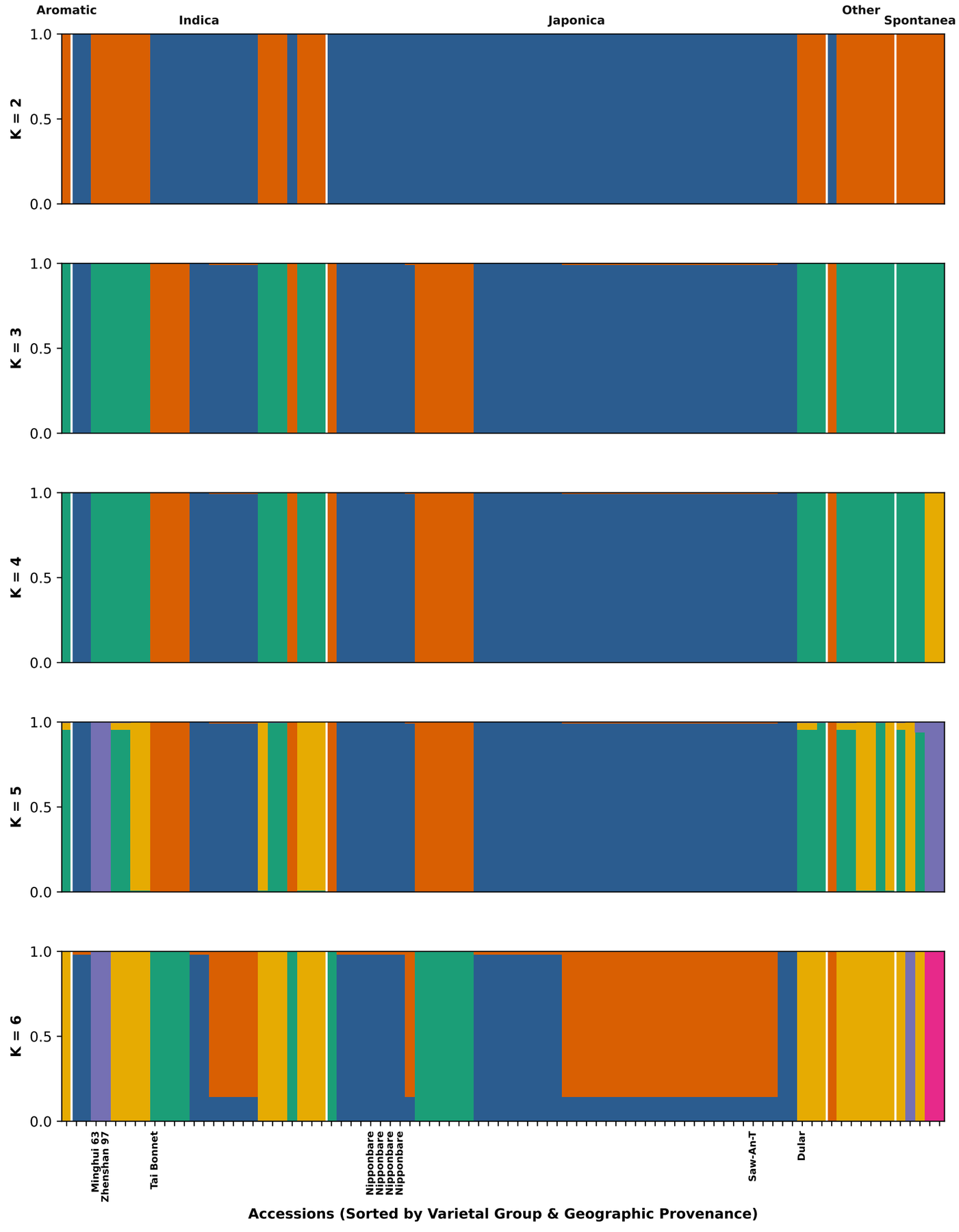
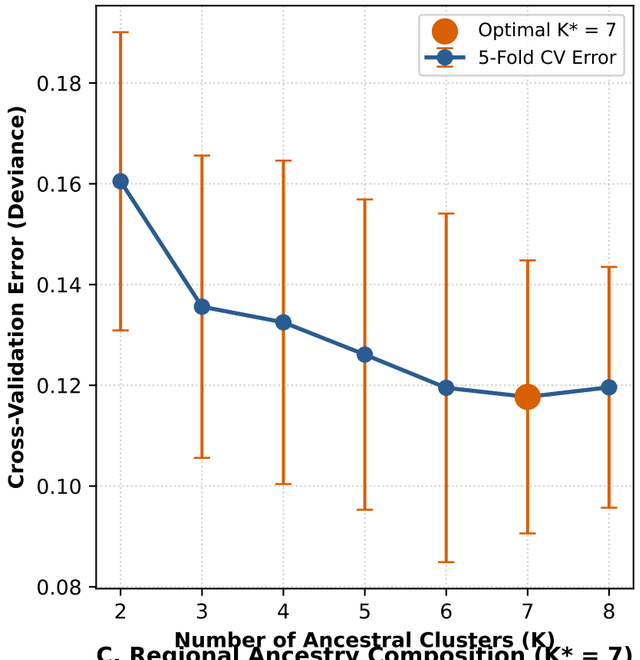
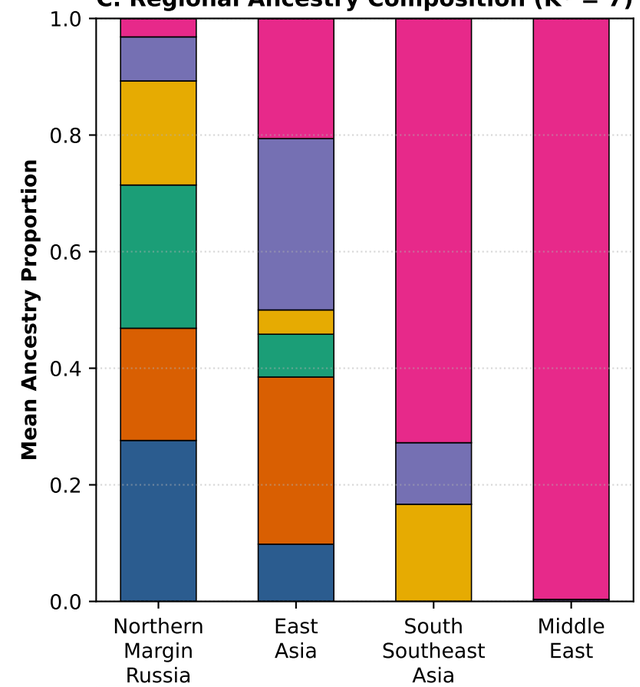
A. Plastome Ancestral Admixture Decomposition Across Global Rice (N = 90)



B. Model Selection: 5-Fold Cross-Validation Curve



C. Regional Ancestry Composition (K* = 7)



D. Ancestral Cluster Architecture (K* = 7) & Model Invariants

Optimal K* = 7 (CV deviance = 0.1177 ± 0.0271) • Evaluated K = 2..8 over 20 tag SNPs

- **C1**: Japonica core
- **C2**: Lowland Indica
- **C3**: Russian Margin
- **C4**: Wild Spontanea
- **C5**: Aus / Aromatic
- **C6**: SSC Isomer
- **C7**: SE Asian Landraces

Model: Bernoulli EM (α = 0.001)